UPI TRANSACTION VOLUME FORECASTING USING TIME SERIES ANALYSIS

Project Documentation

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1. ABSTRACT

The Unified Payments Interface (UPI) has become an important digital payment system in India. Analyzing historical UPI transaction data can help identify growth trends and estimate future transaction activity.

This project focuses on analyzing monthly UPI transaction volume using Time Series Analysis techniques. Moving Average methods and Autoregressive Integrated Moving Average (ARIMA) models are used to study historical patterns and forecast future transaction volumes.

The dataset contains 125 monthly observations covering April 2016 to August 2026. Exploratory Data Analysis, stationarity testing, differencing, ACF and PACF analysis, and ARIMA model development were performed. Three ARIMA models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Among the tested models, ARIMA(2,1,1) achieved the best performance, with a test MAPE of 9.17%. The selected model was used to forecast UPI transaction volume for September 2026 to February 2027. The forecast indicates a potential increase of approximately 10.49% over this period.

2. INTRODUCTION

Digital payments have significantly transformed the financial transaction system in India. UPI enables users to transfer money instantly between bank accounts through mobile applications.

The increasing use of UPI has resulted in substantial growth in monthly transaction volume. Time Series Forecasting provides a way to study historical transaction patterns and estimate future values based on past observations.

This project applies Moving Average and ARIMA techniques to analyze monthly UPI transaction volume and generate short-term forecasts.

3. PROBLEM STATEMENT

UPI transaction activity has grown significantly over the years. Understanding its historical growth and estimating future transaction volumes can provide useful insights into the development of digital payments.

The objective of this project is to analyze historical monthly UPI transaction volume and develop a suitable time series forecasting model to predict future transaction activity.

4. OBJECTIVES

- Analyze historical monthly UPI transaction volume.
- Identify trends and patterns in the dataset.
- Apply Moving Average techniques to smooth fluctuations.
- Test the stationarity of the time series.
- Examine ACF and PACF plots to understand autocorrelation patterns.
- Develop and compare different ARIMA models.
- Evaluate forecasting performance using MAE, RMSE and MAPE.
- Forecast UPI transaction volume for the next six months.

5. DATASET DESCRIPTION

The project uses a monthly UPI transaction dataset.

Dataset details:

- Number of observations: 125
- Time period: April 2016 – August 2026
- Frequency: Monthly
- Target variable: VolumeMn

Important columns:

Column	Description
Date	Monthly observation date
Month	Month name
FY	Financial year
Banks	Number of participating banks
VolumeMn	Transaction volume in millions
ValueCr	Transaction value in crore rupees

The primary target variable used for forecasting is VolumeMn.

6. TOOLS AND TECHNOLOGIES

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Statsmodels
- Scikit-learn

7. METHODOLOGY

The project follows these stages:

1. Data loading and preprocessing.
2. Exploratory Data Analysis.
3. Moving Average calculation.
4. Stationarity testing using the Augmented Dickey-Fuller test.
5. Differencing and ACF/PACF analysis.
6. ARIMA model development.

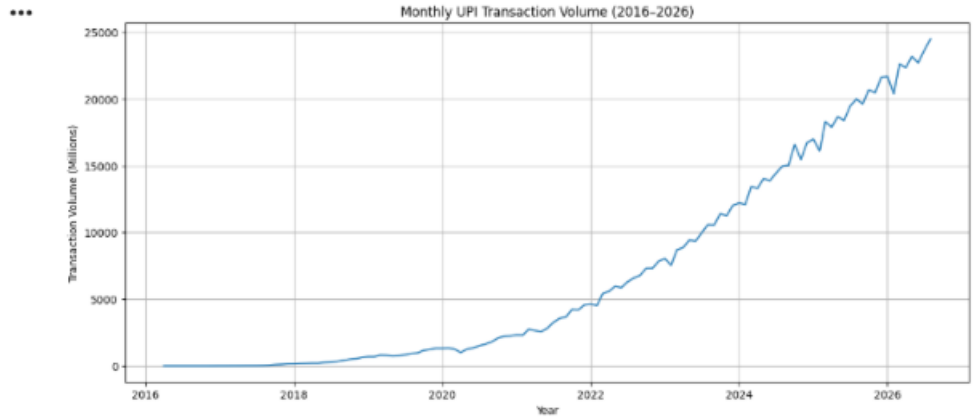
7. Model evaluation and comparison.
8. Residual diagnostics.
9. Future transaction volume forecasting.

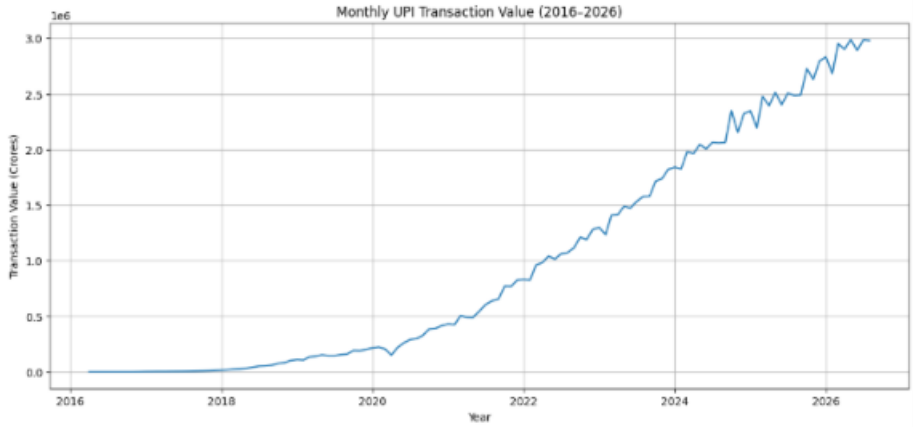
8. EXPLORATORY DATA ANALYSIS

The monthly transaction volume was visualized using line charts.

The analysis revealed a strong upward trend in UPI transaction volume from 2016 to 2026. Transaction activity increased substantially over the observation period, indicating the growing adoption of UPI.

The dataset was also checked for missing values and duplicate dates.





9. MOVING AVERAGE ANALYSIS

Moving Average techniques were applied to smooth fluctuations and identify the underlying trend.

Three rolling averages were calculated:

- 3-Month Moving Average
- 6-Month Moving Average
- 12-Month Moving Average

The 3-month average captures shorter-term changes, while the 6-month and 12-month averages provide smoother representations of the longer-term trend.

The Moving Average analysis showed the overall increase in UPI transaction volume.

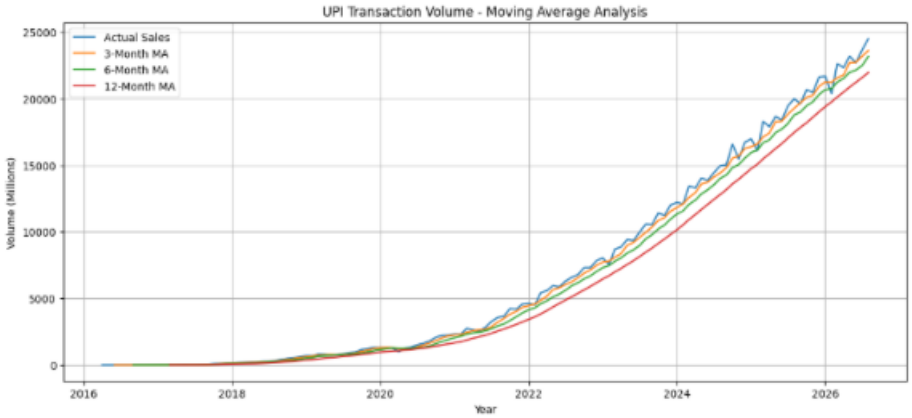


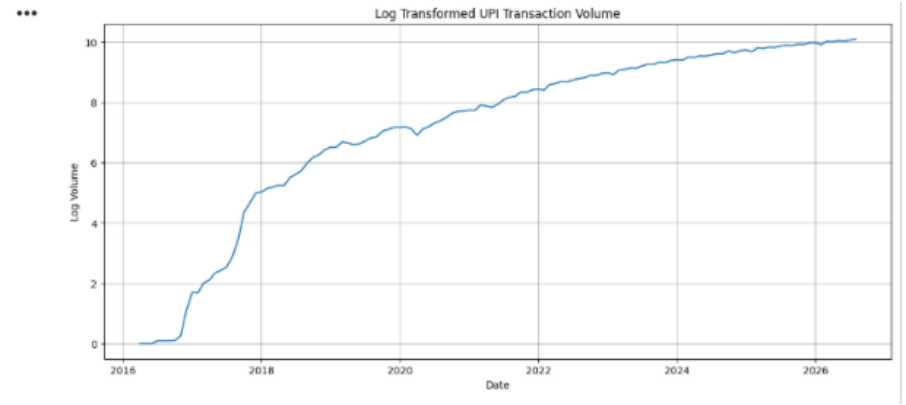
Figure 3: Monthly UPI Transaction Volume – Moving Average Analysis.

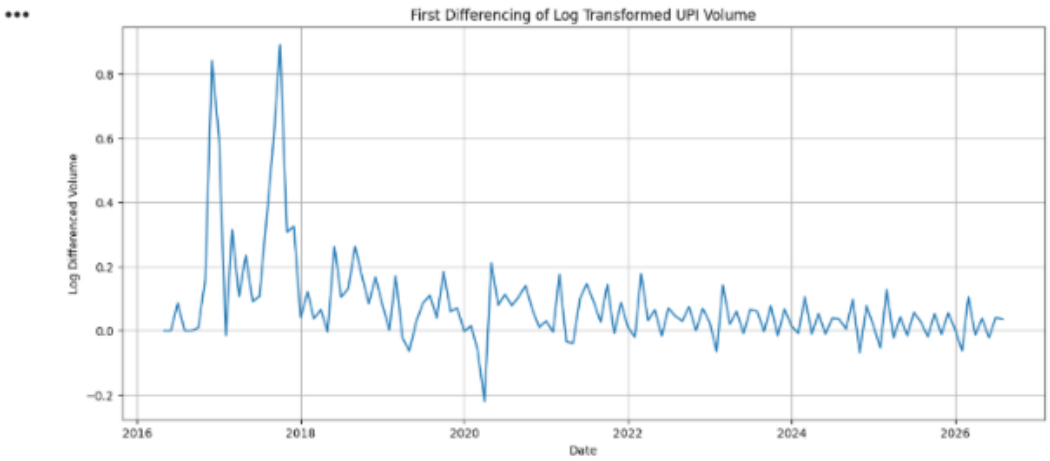
10. STATIONARITY TESTING

The Augmented Dickey-Fuller (ADF) test was used to determine whether the time series was stationary.

The original transaction volume series was non-stationary.

Series	ADF p-value	Result
Original VolumeMn	0.6026	Non-stationary
First Difference	0.8461	Non-stationary
Second Difference	0.000000029	Stationary



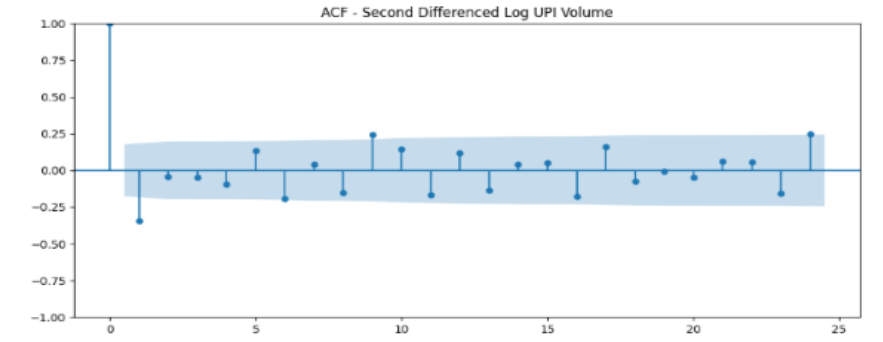


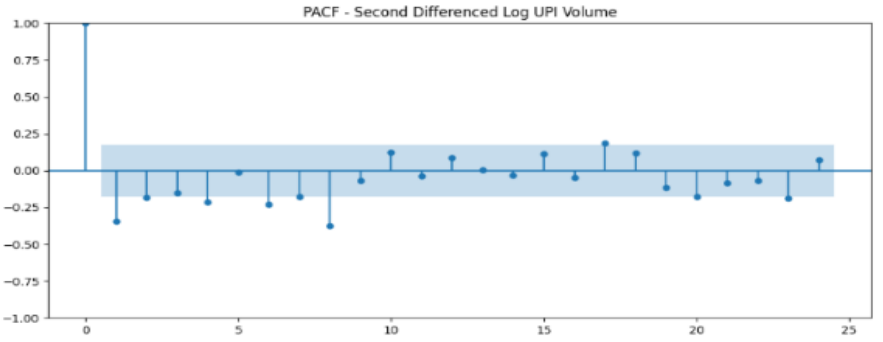
A logarithmic transformation was also explored. The log-transformed series required second differencing to achieve stationarity according to the ADF test.

For the log-transformed series, the ADF p-value was approximately 0.3955 after first differencing and 0.00000725 after second differencing. Therefore, the ADF test indicated stationarity after second differencing for this transformed series.

ACF and PACF plots were examined to understand the autocorrelation structure and guide ARIMA model selection.

Model-selection note: the selected ARIMA(2,1,1) uses d=1, while the ADF result for the log-transformed series points to d=2. ARIMA(2,1,1) was retained because it achieved the lowest held-out test MAE, RMSE and MAPE among the three configurations evaluated. This is a methodological limitation; the differencing order should be validated further before treating the model as definitive.





11. ARIMA MODEL DEVELOPMENT

ARIMA stands for Autoregressive Integrated Moving Average.

It consists of three components:

- AR (p): Autoregressive component.
- I (d): Differencing component.
- MA (q): Moving Average component.

The following models were developed and evaluated:

- ARIMA(3,2,1)
- ARIMA(1,1,1)
- ARIMA(2,1,1)

The models were trained using historical observations and evaluated on a separate test period.

12. MODEL EVALUATION

Three evaluation metrics were used.

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted values.

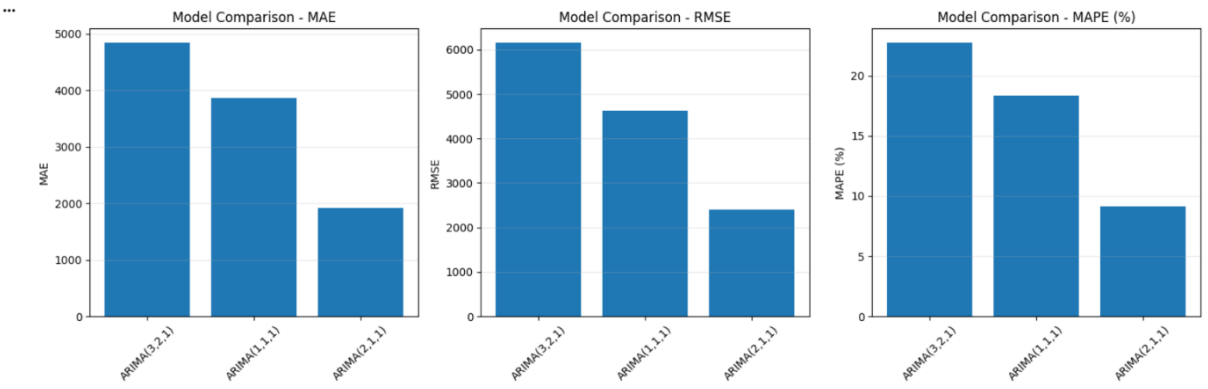
Root Mean Squared Error (RMSE): Measures the square root of the average squared prediction error.

Mean Absolute Percentage Error (MAPE): Measures the average absolute percentage error.

Model Comparison

Model	MAE	RMSE	MAPE
ARIMA(3,2,1)	4,849.78	6,163.63	22.77%
ARIMA(1,1,1)	3,870.25	4,626.43	18.33%
ARIMA(2,1,1)	1,921.79	2,395.08	9.17%

ARIMA(2,1,1) achieved the lowest MAE, RMSE and MAPE among the three evaluated models. Therefore, it was selected for the final forecasting stage.



13. RESIDUAL DIAGNOSTICS

The final model summary reported a Jarque-Bera p-value of 0.00 and a heteroskedasticity p-value of 0.00, indicating non-normal residuals and non-constant variance. These findings should be considered when interpreting the forecasts.

Residual diagnostics were performed on the final ARIMA(2,1,1) model refitted using all 125 monthly observations. The figures below show its residuals over time and the distribution of residual values.

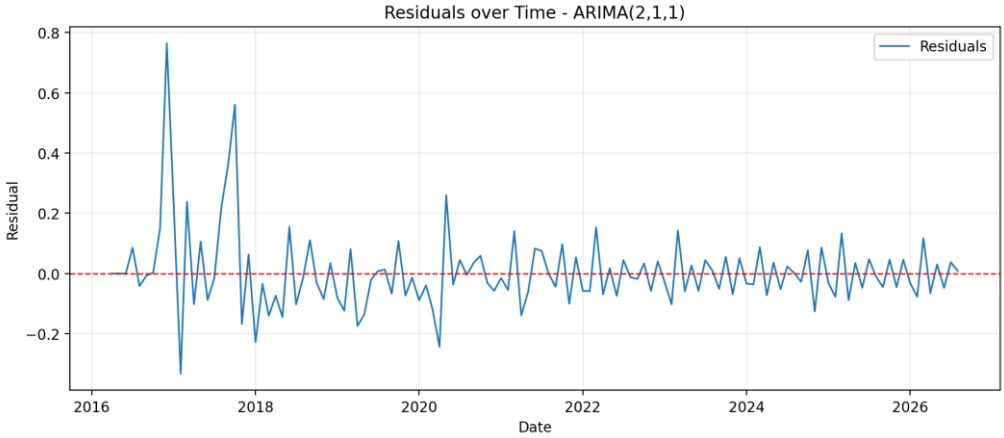


Figure: Residuals over Time – Final ARIMA(2,1,1) Model

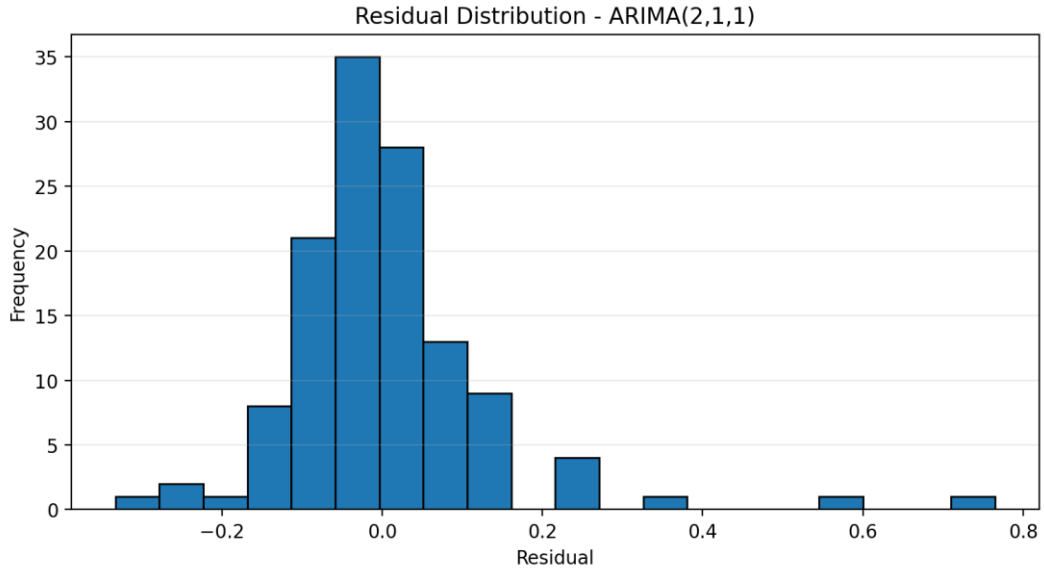


Figure: Residual Histogram – Final ARIMA(2,1,1) Model

In the earlier training-period evaluation of ARIMA(2,1,1), the Ljung-Box p-values at lags 6, 12 and 18 were approximately 0.8421, 0.0775 and 0.1601. All were above 0.05, so that diagnostic did not show statistically significant residual autocorrelation at those selected lags. The final full-data model summary reported a Ljung-Box p-value of 0.97 at lag 1.

However, the final model summary indicated non-normal residuals and heteroskedasticity. These are limitations to consider when interpreting the forecasts.

14. FUTURE FORECASTING

The selected ARIMA(2,1,1) model was refitted using all 125 monthly observations.

A six-month forecast was generated for September 2026 to February 2027.

Forecast Results

Month	Forecasted Volume (Million)
September 2026	25,134.47
October 2026	25,683.20

November 2026	26,208.00
December 2026	26,727.42
January 2027	27,247.93
February 2027	27,771.74

The model predicts a continued increase in monthly UPI transaction volume throughout the forecast period.

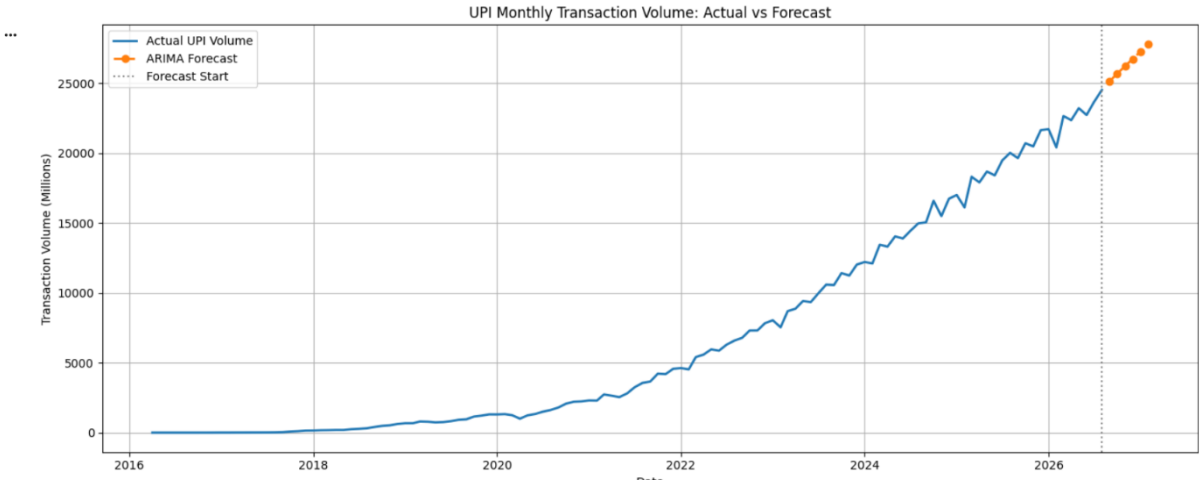
Forecast Growth Analysis

September 2026 forecast: 25,134.47 million.

February 2027 forecast: 27,771.74 million.

Estimated percentage growth: 10.49%.

This indicates a potential increase of approximately 2,637.27 million transactions between the first and last forecast months.



15. KEY FINDINGS

- UPI transaction volume showed a strong upward trend during the observation period.
- Moving Average methods helped smooth short-term fluctuations and identify the underlying trend.
- The original time series was non-stationary.
- Differencing was required to achieve stationarity.
- ARIMA(2,1,1) performed better than the other tested models on the held-out test data.
- The selected model achieved a MAPE of 9.17%.
- The six-month forecast suggests continued growth in UPI transaction volume.
- The forecast estimates a 10.49% increase between September 2026 and February 2027.

16. LIMITATIONS

- The forecast is based on historical transaction patterns and may not capture unexpected changes in UPI usage.

- Only three ARIMA configurations were compared.
- The model residuals were not normally distributed and showed heteroskedasticity.
- The forecast does not explicitly incorporate external factors such as policy changes, economic conditions or changes in digital payment adoption.
- Forecast values are estimates and may differ from actual future transaction volumes.
- Prediction intervals were not included in this analysis.

17. CONCLUSION

This project analyzed monthly UPI transaction volume using Moving Average and ARIMA time series techniques.

Exploratory Data Analysis identified a strong upward trend in transaction activity. Stationarity testing and differencing were used to prepare the data for ARIMA modelling. Multiple ARIMA configurations were developed and compared using MAE, RMSE and MAPE.

ARIMA(2,1,1) achieved the best forecasting performance among the tested models, with a MAPE of 9.17%. The model was subsequently refitted using the complete dataset and used to forecast transaction volume for six months.

The forecast indicates a potential increase from approximately 25,134 million transactions in September 2026 to 27,772 million in February 2027.

Overall, this project demonstrates the application of time series analysis and ARIMA forecasting in understanding historical UPI transaction trends and estimating future transaction activity.

18. FUTURE SCOPE

The project can be extended by:

- Comparing ARIMA with Seasonal ARIMA (SARIMA).
- Evaluating additional ARIMA configurations.
- Including prediction intervals.
- Incorporating external variables that may influence UPI transaction activity.
- Comparing forecasts with actual monthly data as it becomes available.

19. REFERENCES

• Monthly UPI transaction dataset supplied for this project (April 2016 – August 2026).

• Pandas documentation: https://pandas.pydata.org/docs/

• Statsmodels documentation: https://www.statsmodels.org/

• Matplotlib documentation: https://matplotlib.org/stable/

• Scikit-learn documentation: https://scikit-learn.org/stable/